IC improvement round 2

IC improvement round 2: results (2026-10-15)

These are follow-ups to `../ic_improve_20261014/README.md`, run under `RULES.md`:

- Every candidate is compared with the incumbent IC component it replaces.
- Every comparison is paired, uses the same counted unit on both arms, and verifies every logarithm or root set.
- The candidates were tuned on training data, frozen, then tested on held-out data.

#	Candidate vs incumbent	Verdict	Level	Key numbers	Landed
02	Free relations in the production compact-orbit Koblitz IC, K tuned per arm	**ADVANCES**	constant		**Default on** in `examples/koblitz_orbit_dlp_fast_online.rs`. `KIC_FREE_RELATIONS=0` restores the old pipeline byte-identically (A/A 24/24).
04	k = 4 IC: first-dependency stop (d4-ed) vs D_4. This fixes round 1's D_4+2LP regression.	**ADVANCES**	constant (exponent MATCHES)		ecbench base `yline-d4-ed`. Existing hashes are unchanged.
05	Koblitz Gröbner oracle: harvest the multi-variable linear rows that `decisive_rows` discards, root only, gated on generators per variable < 0.9607	**ADVANCES** where the gate fires; MATCHES elsewhere	constant		**Default on** in `koblitz_groebner.rs`. `KIC_LINEAR_HARVEST=0` restores the old engine (counters 60/60).
03	Koblitz Gröbner oracle: gated fall-row augmentation	ADVANCES where it fires; production no-op	constant		Opt-in `KIC_FALLROW_GATE=1`.
01	Genus-3 2LP IC: large primes as columns + structured Gaussian elimination + Wiedemann (`lpsgewd`) vs round-1 IC-2LP (DFS substitution + Lanczos on normal equations)	**ADVANCES**	constant; exponent small		example only
06	Genus-3 2LP IC: frozen κ(q) schedule, block Wiedemann (2,1), and both, vs `lpsgewd` κ = 2	Block Wiedemann **ADVANCES**; schedule and combination **TRADE**	BW: exponent, constant, primitive; schedule: constant		example only

Open follow-ups

- **Koblitz K per size: done (2026-10-15).** `construct:<n>:<a>` with K omitted now uses the tuned K = 125 / 280 / 500 for n = 41 / 53 / 61 (a = 0). An explicit `:<columns>` still wins, so frozen experiments and CI tune grids are unchanged. Checked by an n = 41 solve: `construct:41:0` and `construct:41:0:125` gave identical summary and target records (timings excluded), 8/8 targets solved. Retuning K alone gives 0.46× at n = 41.
- **Online-only Koblitz rows in `docs/ic/LEADERBOARD.md`: kept** (owner decision, 2026-10-15). They are retired only if there is a strong reason.
- **Linear-harvest gate.** Generators per variable is close to (N−1)/N on pipeline systems, so the gate skips the largest gain: ungated root harvest gives 1.28× at (29,15) m2. A gate on variable count or tree size needs its own protocol.
- **Linear harvest end to end at (23,12).** The bench gain (1.17×) does not carry through: the production path stops at the first accepted root, so its solves are shorter than the bench's.
- **Genus-3 linear algebra.** Elimination removes a shrinking share of columns: it keeps 32% of the incumbent's columns at 2^10 but 91% at 2^20. Weight growth turns into column growth, so linear algebra stays near q^1.54 locally. Item 06 tried the κ retune and block Wiedemann. The relation phase now sets the exponent: its local exponent is about 1.43, still about 0.06 above 4/3.

Caveats

- **Host:** shared, 4 vCPU, load average 3–13, isolation level L0. Verdicts use counted operations or thread CPU.
- **Scale:** sizes are small.
- **Conductor:** the control plane was unreachable.